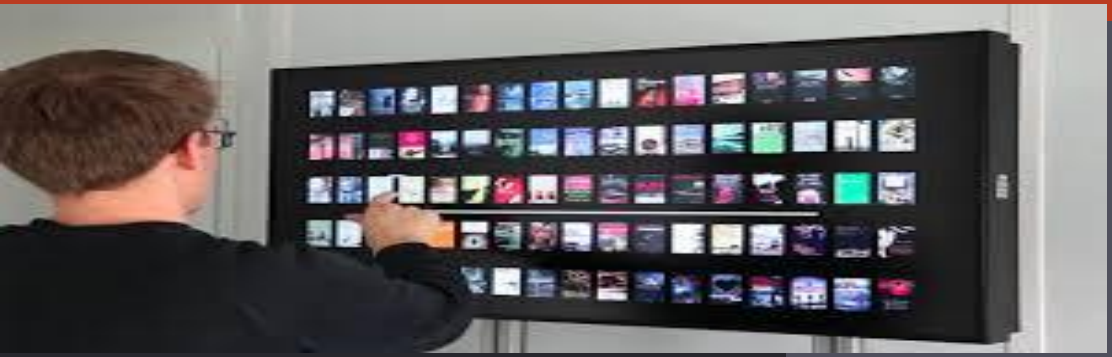
Digital Libraries and Universal Access to Information



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Outline

- What is a Digital Library (DL)?
- Importance of Digital Library
- Why Digital Library?
- Various types of Libraries
- Some Important Digital Libraries
- Conclusion

What is a Digital Library (DL)?

- Digital libraries are a set of electronic resources and associated technical capabilities for creating, searching and using information.
- A digital library is a library in which collection are stored in digital formats and accessible by the computers. The computer contents may be stored in locally. Digital library is a type of information retrieval system.
- The first use of digital library was 1988.

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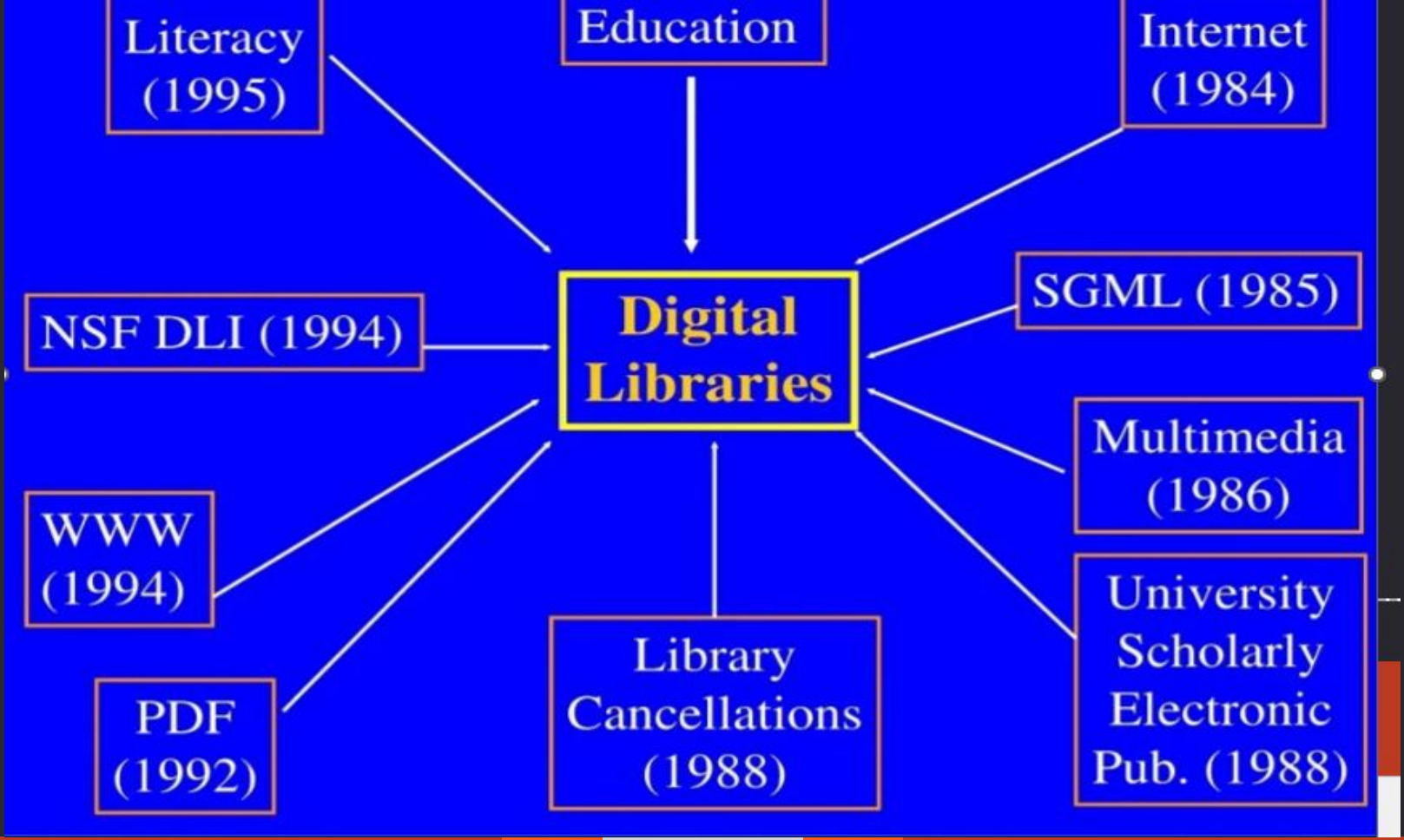
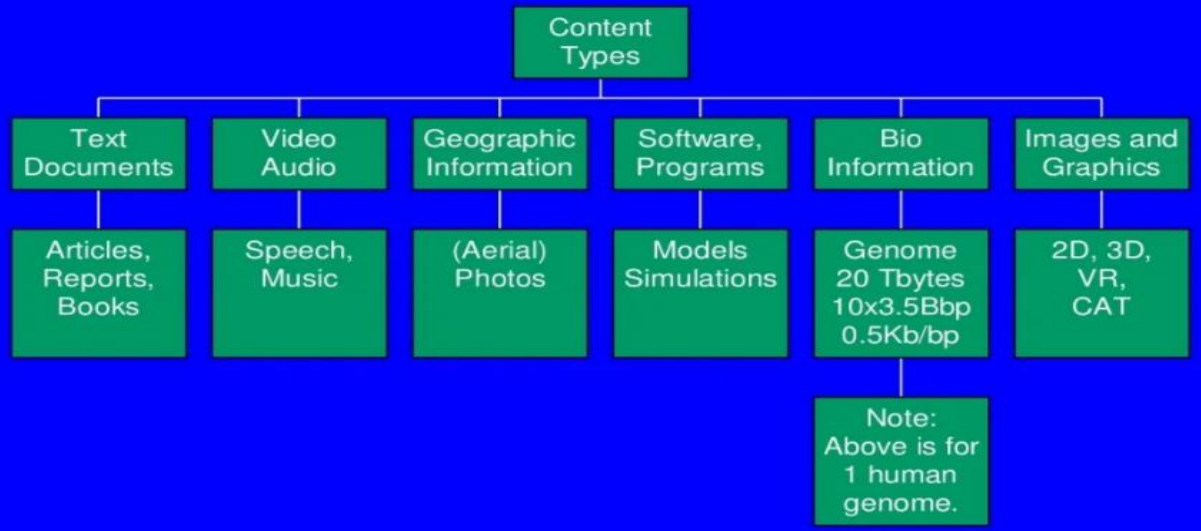
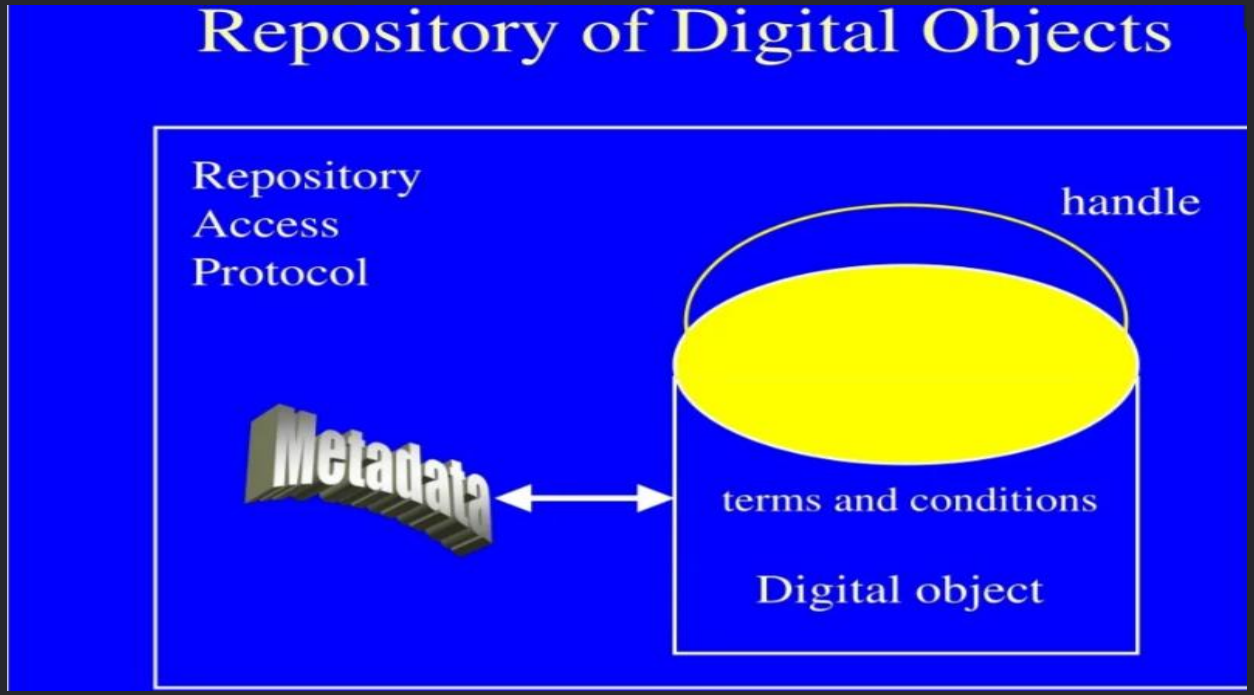
Digital libraries are organized and structured access to information contents in a distributed environment and assist users in searching, evaluating and utilizing resources in different digital formats.


Literacy (1995)
Education
Internet (1984)
NSF DLI (1994)
Digital Libraries
SGML (1985)
Multimedia (1986)
WWW (1994)
University Scholarly Electronic Pub. (1988)
PDF (1992)
Library Cancellations (1988)

Digital Library Content




Repository of Digital Objects
Repository
Access
Protocol
handle
Metadata
terms and conditions
Digital object

5S Layers

- Societies
- Scenarios
- Spaces
- Structures
- Streams

PREVIEW

- Review Classical Definition

Digital libraries are a set of electronic resources and associated technical capabilities for creating, searching and using information.

The content of digital libraries includes data, metadata that describe various aspects of the data (e.g. representation, creator, owner, reproduction rights) and metadata that consist of links or relationships to other data or metadata, whether internal or external]to the digital library. (Borgman 1996)

Definition: 5S Framework

- Societies: interacting people (, computers)
- Scenarios: services, functions, operations, methods
- Spaces: domains + constraints (e.g., distance adjacency): 2D, vector, probability
- Structures: relations, trees, nodes and arcs
- Streams: sequences of items (text, audio, video, network traffic)

Characteristics of Digital Library

- A digital library is a collection of texts, images, or data that has been digitized.
- Copying either by downloading or on-line/ off-line
- It should have a way to navigate and retrieve information.
- It should have at least one specified audience
- The storage of information in digital form
- Usage of communication networks to access and obtain information

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Advantages of the Digital Library

- No physical boundary
- Round the clock availability
- Multiple accesses
- Structured approach
- Information retrieval
- Preservation and conservation
- Space
- Networking and Cost

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Disadvantages of the Digital Library

- Copyright
- Speed of access
- Initial cost is high
- Band width
- Efficiency
- Environment
- Preservation

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Recent Trends

- With the emergence of information society, focus of users changed from container to content and thus information gained importance.
- Role of libraries also changed and it was realized that not only those who are able to read can get benefitted from library but also those who are illiterates can also avail library services through audio , video and other multimedia type of information sources.
- Thus the five laws were modified by making slight changes in the words. Modified forms of the laws are:

- **1. Information is for use.**
- **2. Every user his / her website.**
- **3. Every bit of information its user.**
- **4. Save the time of the reader.**
- **5. The library is a growing organism.**

Library, the changing scenario

- Technological progress has changed how libraries do their work not why. But technology will not substantially alter the business of librarians – connecting people with information
- Librarians may discover that “Libraries without walls” are actually only libraries with new walls – technologically bounded, legally restricted and administratively hamstrung.

Library Today

- Today’s library includes tools with which the user can

  - **Gain access to the holdings of libraries worldwide through automated catalogues.**
  - **Locate both physical and digital versions of scholarly articles and books.**
  - **Optimize searches, simultaneously search the Internet, Commercial databases and library collections.**
  - **Save search results and conduct additional processing to narrow or qualify results.**
  - **From search results, click through to access the digitized content or locate additional terms of interest.**
  - **Customize his/her information request so that the search results reflect individual needs and preferences.**

Evolution of Digital Libraries

1st Generation:

Development of online catalogs

- MARC standard
- ILS systems
- Cataloging cooperatives

2nd Generation:

Experimentation

- Remote access to catalogs
- Abstract/indexing services
- Electronic journals
- Internet use
- Technologies
- "Skunk works" organizations

3rd Generation:

Institutionalization

- Integration of digital material into library collections
- Search for standards
- Benchmark practices
- Modular architectures
- Large collection mgmt.
- User focus
- Integration with library organization

- In the technological environment, different persons have defined these laws differently in order to demonstrate their relevance in the technological dominated world. Gorman (1995) has reinterpreted the five laws as under:

- Libraries serve humanity.
- Respect all forms by which knowledge is communicated.
- Use technology intelligently to enhance service.
- Protect free access to knowledge; and
- Honor the past and create the future.

Universal Access to Information

- Create a Universal Digital Library containing all the books ever published.
- Books in non-native languages remain incomprehensible to most people
  - Translation and Summarization essential for world wide use
  - Current translation systems are not yet perfect
  - Significant improvements in language understanding systems in the past few decades
- Systems based on statistical and linguistic techniques have shown significant performance improvements
  - improve performance using machine learning
- Digitization projects will act as test bed
  - for validating Language Understanding Systems Research
    - e.g. The Million Book Digital Library Project

The Million Book Digital Library

- Collaborative venture among many countries including USA, China and India
- So far 400,000 books have been scanned in China and 200,000 in India
- Content is made freely available around the globe
- Those wishing to see the Video in the next slide should download from

http://www.rr.cs.cmu.edu/MSRI.zip

The Grand Challenge

Create Access to

- All published works online
- Instantly available
- In any language
- Anywhere in the world
- Searchable, browsable, navigable
- By humans <u>and</u> machines

One Step at a Time…

- Million Book DL
  - Only about 10% of all the world's books
    - Harvard University 12M
    - Library of Congress 30M
    - OCLC catalog 42M
    - All Multilingual Books ~100M
- At the rate of digitization of the last decade it would take a 100 years!

Million Book Project: Issues

- Time
  - At one page per second (20,000 pages per day shift), it will take 100 years (200 working days per year) to scan a million books of 400 pages each
- Cost
  - 100M books at US$100 per book would coat $10B
  - Even in India and China the cost will be $1B
  - The annual cost is currently expected to be close $10M per year with support from US, India and China.
- Selection
  - Selection of appropriate books for scanning is time consuming and expensive

Million Book Project: Issues (cont)

- Logistics
  - Each containers hold 10,000 to 20,000 books. Shipping and handling costs about $10,000
- Meta Data
  - Accessing and/or creating Meta data requires professionals trained in Library science
- Optical Character Recognition Technology
  - Essential for searching, translation and summarization
  - Many languages don’t have OCR

Million Book Project: Status

- 21 Centers in India
- 17 centers in China
- 1 Center in Egypt
- Planned : Australia and Europe
- About 600,000 books scanned
  - About 120,000+ accessible on the web from India
    - http://dli.iiit.ac.in/
    - Uses 8TB of storage
  - 10 TB server at CMU Library planned for 2015
  - 1,000,000 books by the end of 2014
  - Capacity to scan a million pages a day expected to be operational by the end of 2014

IMG_2038.MOV

HathiTrust

Universal Digital Library

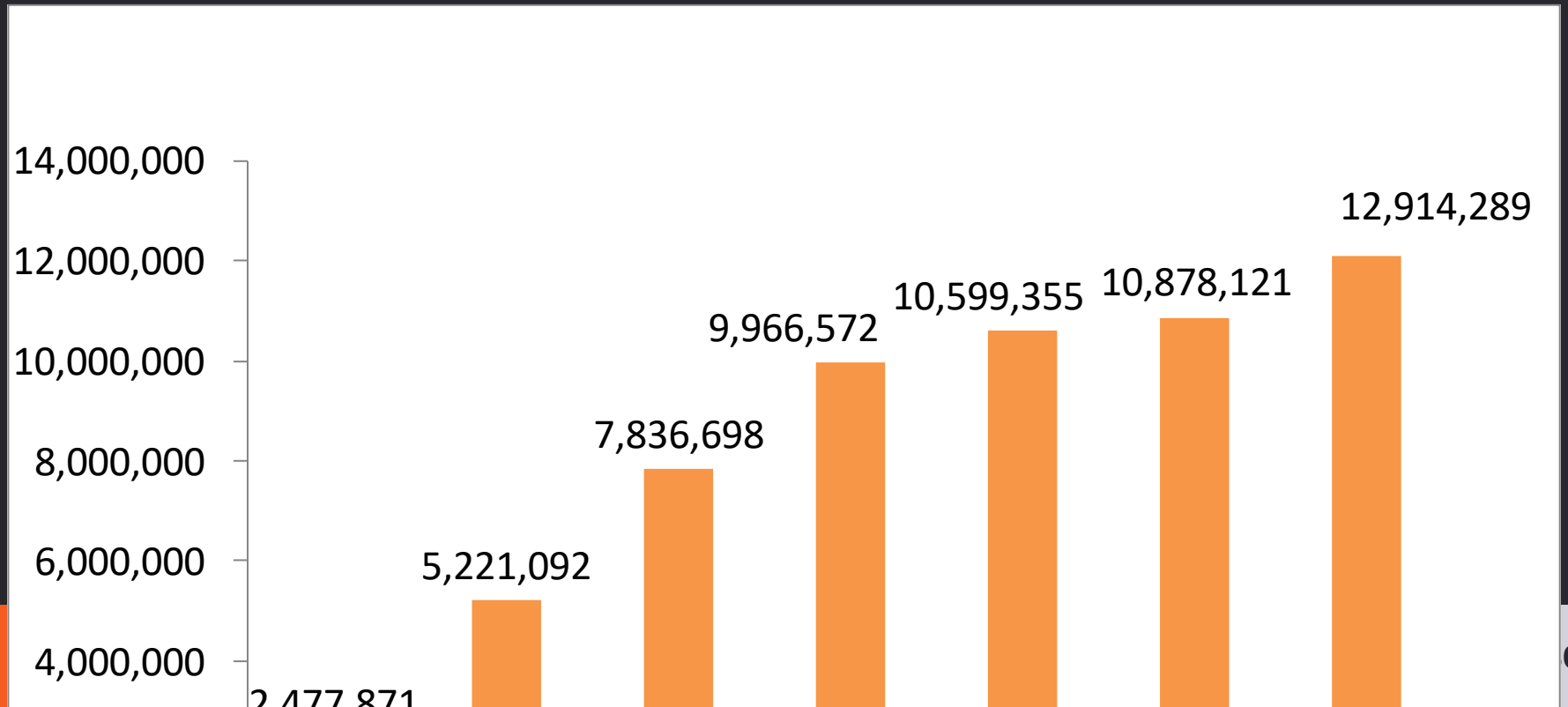
Common Goal

Single Entity, Many Partners

Current Partners

- Columbia University
- New York Public Library
- University of California system
- CIC (Committee on Institutional Cooperation)

University of Chicago
University of Illinois
Indiana University
University of Iowa
University of Michigan
Michigan State University
University of Minnesota
Northwestern University
Ohio State University
Pennsylvania State University
Purdue University
University of Wisconsin-Madison


14,000,000
12,000,000
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8,000,000
6,000,000
4,000,000
2,477,871
5,221,092
7,836,698
9,966,572
10,599,355
10,878,121
12,914,289

Focus on users

- Preservation…with Access
- Brings concerns of research libraries to bear on the way the scholarly record is cared for and made available
  - Scholarly Resource
  - Bibliographic Search
  - Full-text search
  - Collections
  - Full-PDF download of public domain

Digital Library of Institutional Repositories

An Institutional Repository (IR) is a digital collection or archives of a university's intellectual output. Universities and other institutions are producing digital information base of their Ph.D. theses& dissertations, articles, reports, conference proceedings, lecture notes, presentational, audio-video records using open source software and making them available to their end users. There are 92 institutional and subject wise repositories have been registered in Registry of Open Access Repository (ROAR). Institutional repositories initiated in India can be viewed at http://roar. eprints.org/

Cornell University Library

DSpace Digital Repository

Search DSpace:

Go

Advanced Search

Home

Browse

Communities & Collections

Titles

Authors

By Date

Sign on to:

Receive email updates

My DSpace

authorized users

DSpace at Cornell University >

Internet-First University Press

Community home page

In: Internet-First University Press

Search for Go

or browse Titles Authors By Date

Internet First University Press

These materials are being published as part of a new approach to scholarly publishing. The manuscripts and videos are freely available from this Internet-First University Press repository within DSpace at Cornell University.

Recent Submissions

Immigrants and the Community: Farmworkers with Families

Open Scholarship and Research Universities

The Visible Hand, Volume 12, Issue 2

Contested Values: Democracy and Diversity in American Culture

New Concepts of Cattle Growth

ETD at Indian Institute of Science: Home - Mozilla

File Edit View Go Bookmarks Tools Window Help

Back Forward Reload Stop http://etd.ncsi.iisc.ernet.in/ Search Print

etd@IISc

Electronic Theses and Dissertations of Indian Institute of Science

Search ETD: Go

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Home

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Communities & Collections

Titles

Authors

Guides

By Date

By Subject

Sign on to

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Edit Profile

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Help

Project Site

Study@IISc

ETD at Indian Institute of Science >

Welcome to etd@IISc!

This is the digital repository of Theses and Dissertations of Indian Institute of Science, Bangalore, India. You can search, browse and access theses and dissertaions from this collection. This repository has been developed to capture, disseminate and preserve research theses of Indian Institute of Science. It complements ePrints@IISc, the research publications repository of IISc.

Number of theses in etd@IISc: 92

Search

Enter some keywords or text in the box below to search the whole Repository (See Help for more details)

Go

Browse communities and collections in etd@IISc

Division of Biological Sciences [18]

Division of Chemical Sciences [12]

Division of Electrical Sciences [38]

Division of Information Sciences [3]

Division of Mechanical Sciences [17]

Division of Physical and Mathematical Sciences [4]

September 12, 2005

ETD Help

How to Submit a thesis

Thesis Templates

National & International ETD Services

NDLTD

Search Thesis World wide

Vidyanidhi

etd@IISc is a joint service of NCSI & IISc Library || Contact:admin@etd.ncsi.iisc.ernet.in

Powered by DSpace || Compliant to OAI-PMH V 2.0 and ETD-MS V 1.01

World Digital Library

- The **World Digital Library** (**WDL**) is an international digital library operated by UNESCO and the United States Library of Congress.
- The WDL has stated that its mission is to promote international and intercultural understanding, expand the volume and variety of cultural content on the Internet, provide resources for educators, scholars, and general audiences, and to build capacity in partner institutions to narrow the digital divide within and among countries.

Digital Divide

- ICTs are an issue of contention
  - Technology both facilitates as well as restricts access to information
  - democratization of access to knowledge and information should be the focus
- DLs are among the most significant offshoots of the Internet and web technologies and should be seen as important mechanisms in bridging the digital divide

Digital Library Technology is the key to

- Preserving this multiple layers of resources (and Knowledge)
- Making the knowledge accessible
- Making the ordinary citizen an effective participant in the proper functioning of democracy and democratic institutions
  - Access to information is essential and underpins other human rights
  - on-line digital environment provides a powerful arena for people

National Policy on Digital Library, INDIA

The National Task Force on IT and Software Development (2003) has given some valuable recommendations for development of DLs in the country related to development of pilot project on Digital Library, preservation of vast wealth of Indian traditional knowledge, digitization of Indian Theses and Dissertations and copyright protection. However, there is no any clear cut national policy on Digital Libraries.

National Digital Library of India

The National Digital library of India (NDLI) is a project under Ministry of Human Resource Development, The Government of India.

The objective is to collect and collate metadata and provide full text index from several national and international digital libraries, as well as other relevant sources.

It is a digital repository containing textbooks, articles, videos, audio books, lectures, simulations, fiction and all other kinds of learning media. The NDLI provides free of cost access to many books in the Indian languages and English.

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Current Digital Library Initiatives in India

Government institutions of national importance, national level institutions, research organization, universities, state government institutions, financial institutions, private institutions are key player for initiating digital libraries in India. Some of the important digital library initiatives and programmes initiated across the country are as follow:

Digital Library of India (http://www.dli.ernet.in/)

Digital Library of India was formally launched by the then president of India Dr. A. P J. Abdul Kalam in 8th September 2003 to preserve knowledge and cultural heritage of India.

It is part of Universal Digital Library Project of the US-NSF and Million Books Project envisaged by Carnegie Mellon University, USA.

The project is supported by Ministry of Communications and Information Technology, Government of India and coordinated by Indian Institute of Science, Bangalore.

It digitizes and preserves all significant library, artistic and scientific works in its three regional mega scanning centre and 21 scanning centers and makes it freely available to world for education and research.

VigyanPrasar Digital Library (http://www.vigyanprasar.gov.in/digilib/)

VigyanPrasar, an autonomous organization under Department of Science and Technology, Government of India, was established in 1989 for communication of science and technology.

VigyanPrasar maintains an open access digital library to spread scientific knowledge, where the digital collection contains digitized full-text versions of all significant scientific works that are being published .

Digital Library of Manuscripts

Kalasampada: Digital Library Resources for Indian Cultural Heritage (http://www.ignca.nic.in/dlrich.html) The Indira Gandhi National Centre for the Arts (IGNCA) in collaboration with the Ministry of Communication and Information Technology initiated a digital library on indigenous cultural heritage named Kalasampada (Digital Library: Resources of Indian Cultural Heritage (DL-RICH)) for the development of a databank of cultural heritage.

National Mission for Manuscripts (http://www.namami.org/index.htm)

The Department of Culture, and Ministry of Tourism and Culture, Government of India, launched the National Mission for Manuscripts in February 2003 with an aim to locate, document, preserve and render vast collection of manuscripts which are available in variety of themes, textures and aesthetics, scripts, languages, calligraphies, illuminations and illustrations.

National Digital Library of Electronic Thesis & Dissertation

Shodhganga is a national repository of electronic theses and dissertation. It was launched on 20th May 2010 by INFLIBNET Centre under support of University Grants Commission with an aim to facilitate open access to Indian theses and dissertations to the academic community world-wide.

Digital Library of Journals-Initiatives by Scientific Society and Publishers

Indian Academy of Sciences (www.ias.ac.in/pubs/journals/) The Indian Academy of Sciences (IAS) is a scientific academy funded by the Government of India.

Digital platform is developed to provide online access to 11 peer reviewed journals with all backfiles and other publications including reports, newsletter, patrika, year book and annual report etc. published by Indian Academy of Science. It provides access to 11 journals with back files.

Indian National Science Academy (www.insa.ac.in) The Indian National Science Academy (INSA) was established in 1935.

It is funded by Government of India. INSA, under the support of NISSAT, has initiated e-journal@insa project in 2002 to facilitate conversion of INSA journals from print to digital format and host these materials online. This portal provides access to current and back volume full-text literature of INSA journals,

Impact of digital library research?

- User studies
  - How much do we know about changing patterns of research, learning and engagement?
- Federation and metasearch
  - FDI, IndexData, Cheshire, iPort, …
  - OAI/OpenURL
  - NISO metasearch – issues still to be addressed
- Repositories/digital library systems
  - Multiple communities
  - Dspace, Fedora, CONTENTdm, DLXS, ..
- Metadata
  - Growing acronymic density
  - Collections, rights, policies, services, …
  - Complex objects, relations
- Identifiers/citation

Local successes …

… but we have many open questions.

Technical Issues

- High bandwidth network facilities supporting multimedia document transfer
- Open communication protocols (Z39.50 protocol)
- Information access tools (browsers, display and search tools)
- Meta-databases
- Electronic Publishing tools
- Data compression
- Digital storage devices
- Scanning and conversion technologies

China – Visit

M.Krishnamurthy

Technical Issues

- Media integration technology (multimedia)
- Advanced retrieval, indexing, and filtering etc.,
- Document description and representation standards (SGML)
- Interoperability over the network
- Privacy, authentication and security of information
- Location dependent naming of digital source
- Licensing agreements

Challenges

1. Interoperability
   Technical, process, language, (Different platforms etc)
2. Building digital collections
   - *2.a* Digitization
   - 2.b Acquisition of original digital works
   - 2.c Access to external materials
3. Metadata
4. Naming, identifiers, and persistence
   - 4.a Uniform Resource Locator (URL)
   - 4.b Persistent Uniform Resource Locator (PURL)
   - 4.c Digital Object Identifier

Conclusion

Digital libraries provide a variety of benefits that can help users in many ways. Digital library existence certainly greatly helps educational academics in finding learning resources.

The core benefits of digital libraries are easy access, unlimited space and time, preservation of collections.

The position of the digital library is complementary to the existence of conventional libraries in facing technological developments.

From many benefits provided by digital libraries, do not makes it replace the position of conventional libraries

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Be not afraid of growing slowly, be only afraid of standing still”.

Progress is **promising**..

Thanks!